

thrFiringSchmitt

1 Executive Summary

A Schmitt trigger logic is implemented to map a desired thruster force value into a thruster on command time.

The module reads in the attitude control thruster force values for both on- and off-pulsing scenarios, and then maps this into a time which specifies how long a thruster should be on. The thruster configuration data is read in through a separate input message in the reset method. The Schmitt trigger allows for an upper and lower bound where the thruster is either turned on or off. Commands that saturate the control period are over-driven by a configurable `onTimeSaturationFactor` (default 1.0) to guarantee thrust for the full duration of the control period. The paper Steady-State Attitude and Control Effort Sensitivity Analysis of Discretized Thruster Implementations includes a detailed discussion on the Schmitt Trigger and compares it to other thruster firing methods.

2 Message Connection Descriptions

The following table lists all the module input and output messages. The module msg connection is set by the user from python. The msg type contains a link to the message structure definition, while the description provides information on what this message is used for.

Table 1: Module I/O Messages

Msg Variable Name	Msg Type	Description
thrConfInMsg	THRArrayConfigMsgF32Payload	Read in <code>reset()</code> . Contains <code>numThrusters</code> and per-thruster max thrust.
thrForceInMsg	THRArrayCmdForceMsgF32Payload	Read every <code>updateState()</code> . Provides commanded forces F_i . Values may be negative in <code>OFF_PULSING</code> mode; negative forces are clamped to zero after the pulsing adjustment.
onTimeOutMsg	THRArrayOnTimeCmdMsgF32Payload	Thruster on-time requests t_i . Values are non-negative, and may be set to <code>onTimeSaturationFactor * control period</code> when saturation is detected.

3 Module Parameters

The following table lists all the module parameters than can be set. The parameters are optional unless indicated (if not specified default is used).

Table 2: Module Parameters

Parameter Name	Type	Units	Default	Description	Bounds
levelOn, levelOff (required, set together via <code>setLevelsOnOff</code>)	float, float	[-]	0, 0	ON and OFF duty cycle fractions	$0.0 < \text{levelOn} \leq 1.0$, $0.0 \leq \text{levelOff} < 1.0$, $\text{levelOn} \geq \text{levelOff}$ (checked in setter)

Parameter Name	Type	Units	Default	Description	Bounds
thrMinFireTime (required)	float	[s]	0	Minimum ON time for thrusters	Must be greater than zero (checked in setter)
thrustPulsingRegime	enum ThrustPulsingRegime	[-]	ON_PULSING (0)	Gates on-pulsing (ON_PULSING) or off-pulsing (OFF_PULSING)	N/A
controlPeriod (required)	float	[s]	0	Control period (time between two update calls, i.e. $1/\text{fsw_rate}$)	Must be greater than zero (checked in setter)
onTimeSaturationFactor	float	[-]	1.0	Factor applied to control period when on-time saturates	Must be ≥ 1.0 (checked in setter)

The `setLevelsOnOff` setter validates that `levelOn` is greater than or equal to `levelOff`.

4 Module Assumptions and Limitations

- The incoming forces F_i can be both positive or negative, depending if an on- or off-pulsing mode is being implemented. The particular mode is set through `thrustPulsingRegime`.
- It is assumed that `thrMinFireTime` is less than the control period.
- Max thrust is taken from configuration; must be ≥ 0 per thruster (negative values throw).

5 Initialization

The module is configured by:

```

module = thrFiringSchmittF32.ThrFiringSchmitt()
module.modelTag = "thrFiringSchmitt"
module.setLevelsOnOff(0.75, 0.25)
module.thrMinFireTime = 0.02
module.thrustPulsingRegime = thrFiringSchmittF32.ThrustPulsingRegime_ON_PULSING
module.controlPeriod = 0.5
module.onTimeSaturationFactor = 1.1

```

6 Detailed Module Description

This module implements a Schmitt trigger thruster firing logic. The minimum desired on-time $\Delta t_{\min}$ is specified by the user. If the commanded on-time exceeds this minimum, the thruster fires for the computed duration. If the on-time falls below the minimum, the Schmitt trigger hysteresis logic determines whether the thruster should fire for the minimum duration or not fire at all.

For thruster i with commanded force F_i , maximum thrust $F_{\max,i}$, control period Δt , and minimum on-time $\Delta t_{\min}$:

1. OFF_PULSING adjustment ($F_i \leq 0$):

$$F_i = \begin{cases} F_i + F_{\max,i}, & \text{if OFF_PULSING} \\ F_i, & \text{if ON_PULSING} \end{cases}$$

2. Clamp negative force:

$$F_i = \max(F_i, 0)$$

3. Nominal on-time from force request:

$$t_i = \frac{F_i}{F_{\max,i}} \Delta t$$

4. Schmitt trigger logic (if $t_i < \Delta t_{\min}$):

Compute the duty cycle $l = t_i / \Delta t_{\min}$.

- If $l \geq l_{\text{on}}$: turn thruster on, $t_i = \Delta t_{\min}$
- If $l \leq l_{\text{off}}$: turn thruster off, $t_i = 0$
- Otherwise: maintain previous state (on $\rightarrow \Delta t_{\min}$, off $\rightarrow 0$)

5. Normal range (if $\Delta t_{\min} \leq t_i < \Delta t$):

t_i is used as-is. Thruster state is set to on.

6. Saturation (if $t_i \geq \Delta t$):

$$t_i = f_{\text{sat}} \Delta t$$

where f_{sat} is the `onTimeSaturationFactor` (default 1.0).

The output stores t_i in `onTimeRequest[i]`.